



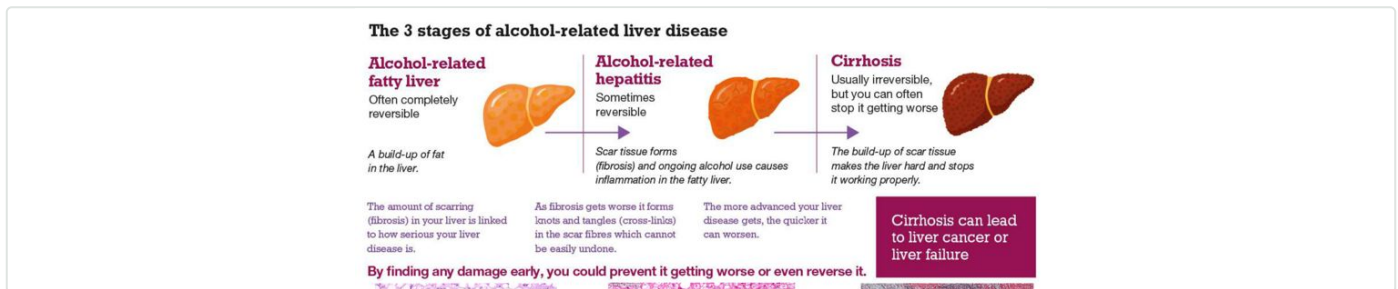
BEFORE YOU READ — ANSWER FROM MEMORY

1. What proportion of chronic heavy drinkers develop each stage: steatosis, steatohepatitis, cirrhosis?
2. Which gene and chromosome are mutated in hemochromatosis vs Wilson disease?
3. Why does α 1AT deficiency cause liver disease by one mechanism but lung disease by a completely different one?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Alcohol-related liver disease

Stage	Frequency	Mechanism
Hepatic steatosis	90-100% of chronic excessive drinkers	Ethanol increases lipid synthesis and impairs lipid excretion → intracellular fat accumulation
Steatohepatitis	10-35% , sometimes after weeks-months	Further fat accumulation + toxic ethanol effect → inflammation with neutrophilic infiltration
Fibrosis and cirrhosis	8-20% , usually after 15-20 years	Ongoing cell destruction and inflammation → fibrosis → cirrhosis



The 3 stages of alcohol-related liver disease: fatty liver (reversible) → hepatitis (sometimes reversible) → cirrhosis (usually irreversible)

Significance	A major cause of chronic liver disease and cirrhosis-related death in Western countries.
HCC risk	10-20% of patients with alcohol-related cirrhosis develop hepatocellular carcinoma.
Regression	Complete abstinence, even with established cirrhosis, may cause partial regression of scarring and regeneration from micronodular to macronodular cirrhosis; complete regression is rare .

2 Nonalcoholic fatty liver disease (NAFLD)

What	Fatty liver disease associated with insulin resistance and the metabolic syndrome .
Associations	Type 2 DM (or family history) · central obesity · dyslipidemia (high TG, low HDL, high LDL) · hypertension.
Spectrum	Can show the same three changes as alcohol-related liver disease: steatosis, steatohepatitis, cirrhosis.
NASH	"Nonalcoholic steatohepatitis" : overt liver injury (elevated transaminases) with histologic hepatitis; progression from steatosis → NASH → cirrhosis is low frequency but NASH can lead to hepatocellular carcinoma .

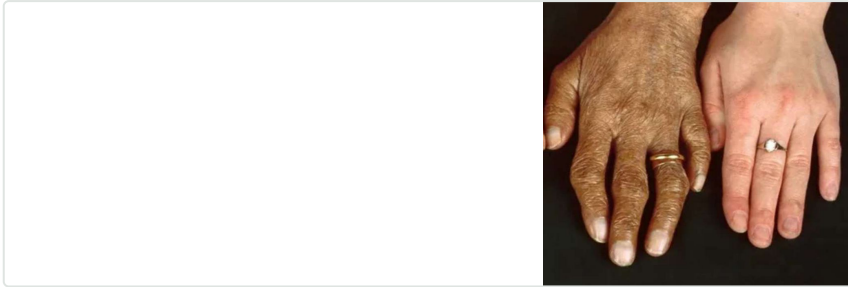
3 Inherited metabolic liver diseases

Hemochromatosis

What	Increased iron levels deposit in tissues, causing injury.
Hereditary (primary)	Most common form: autosomal recessive HFE gene (chromosome 6p) → increased small-intestinal iron absorption.
Acquired (secondary)	Follows blood transfusions for chronic anemias.
Sex distribution	5× more common in males .

Mechanism of injury	Detail
Lipid peroxidation	Iron-catalyzed free radical reactions
Fibrosis	Stimulation of collagen formation via hepatic stellate cell activation
DNA damage	Reactive oxygen species → lethal cell injury or predisposition to HCC
Tissue	Effect
Liver	Micronodular cirrhosis and hepatocellular carcinoma
Skin	Hyperpigmentation (“bronzing”)
Endocrine glands	Secondary diabetes mellitus
Heart	Heart failure and arrhythmias
Gonads	Hypogonadism

Diagnosis: markedly elevated serum iron and ferritin; increased tissue iron on liver biopsy with **Prussian blue** stain.

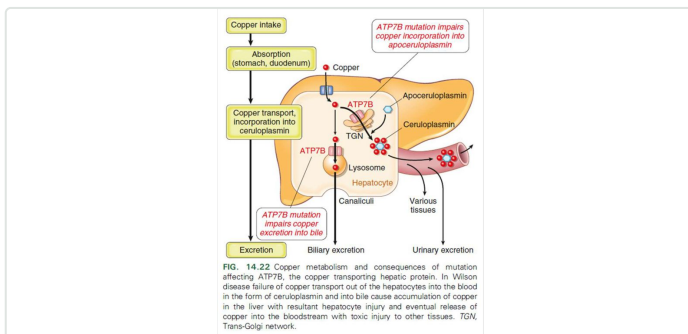


Hemochromatosis: skin bronzing (left hand) compared with a normal hand

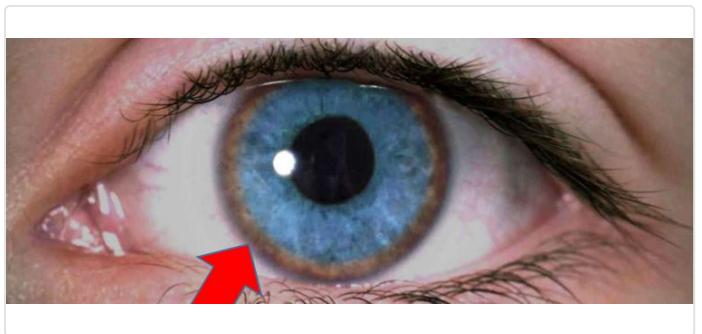
Wilson disease

What	Autosomal recessive, loss-of-function mutation of ATP7B (chromosome 13) → impaired biliary copper excretion and failure to incorporate copper into ceruloplasmin.
Where copper deposits	Brain, corneas, kidneys, bones, joints, parathyroid glands.
Organ	Damage
Liver	Mild-moderate steatosis → acute fulminant hepatitis → chronic hepatitis → steatohepatitis → cirrhosis (variable spectrum)
Brain	Primarily the basal ganglia
Eyes	Kayser-Fleischer rings: green-brown copper deposits in Descemet membrane at the corneal limbus

Diagnosis: low serum ceruloplasmin · increased hepatic copper (>250 µg/g dry liver weight is diagnostic) · increased urinary copper excretion. **Serum copper is not diagnostic** (may be low, normal, or high). Genetic diagnosis is difficult given the many possible ATP7B mutations.



Copper metabolism and the consequence of ATP7B mutation in Wilson disease



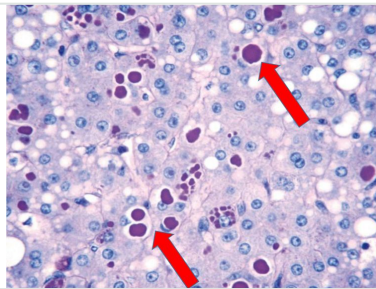
Kayser-Fleischer ring: copper deposition at the corneal limbus

Alpha-1-antitrypsin (α1AT) deficiency

What	Autosomal recessive mutation (chromosome 14) → misfolded, nonfunctional α1AT.
Normal role of α1AT	Inhibits proteases released by neutrophils at sites of inflammation; deficiency → unopposed protease activation → tissue destruction.

Organ	Mechanism & effect
Liver	Abnormal α 1AT accumulates within hepatocytes → ER stress → hepatocyte apoptosis; damage ranges from steatosis to hepatitis to cirrhosis
Lungs	Unopposed neutrophil protease (elastase) activity destroys alveolar elastic fibers → pulmonary emphysema

Liver biopsy: **PAS-positive, diastase-resistant globules** within hepatocytes.



α 1AT deficiency: PAS-D positive globules within hepatocytes (arrows)

P Must-remember pearls

- **Alcohol-related liver disease progresses steatosis (90-100%, reversible) → steatohepatitis (10-35%) → fibrosis/cirrhosis (8-20%, usually irreversible)**; 10-20% of cirrhotics develop HCC.
- **NAFLD = fatty liver + insulin resistance/metabolic syndrome** (T2DM, central obesity, dyslipidemia, hypertension); can progress through the same steatosis→NASH→cirrhosis spectrum as alcohol.
- **Hemochromatosis: HFE gene, chr 6, autosomal recessive, 5× more common in males** → micronodular cirrhosis + HCC, skin bronzing, secondary DM, cardiomyopathy, hypogonadism; diagnosed by iron studies + **Prussian blue** stain.
- **Wilson disease: ATP7B gene, chr 13** → low ceruloplasmin, high hepatic/urinary copper, **Kayser-Fleischer rings**, basal ganglia damage; serum copper is **not** diagnostic.
- **α 1AT deficiency** damages the liver by **accumulation** of misfolded protein (PAS-D positive globules, ER stress → apoptosis) but damages the lung by a completely different mechanism: **loss of antiprotease activity** → unopposed elastase → emphysema.

Q Test yourself

1. A patient with metabolic syndrome (type 2 diabetes, central obesity, dyslipidemia) is found to have hepatic steatosis on ultrasound, with no history of alcohol use. The most likely diagnosis is:

- A. Nonalcoholic fatty liver disease
- B. Alcohol-related fatty liver
- C. Wilson disease
- D. Hemochromatosis
- E. Autoimmune hepatitis

2. A 45-year-old male presents with bronze skin pigmentation, new-onset diabetes, and cirrhosis. Which stain on liver biopsy would support the diagnosis?

- A. PAS with diastase
- B. Congo red
- C. Rhodanine (copper stain)
- D. Prussian blue
- E. Oil red O

3. A young patient with liver disease, dysarthria, and behavioral changes is found to have a Kayser-Fleischer ring on slit-lamp exam and low serum ceruloplasmin. Which gene is most likely mutated?

- A. HFE
- B. SERPINA1 (α 1AT gene)
- C. ATP7B
- D. APC
- E. PRNP

4. Liver biopsy from a child with early cirrhosis and a sibling with emphysema shows PAS-positive, diastase-resistant globules within hepatocytes. This is most consistent with:

- A.** Hemochromatosis
- B.** Wilson disease
- C.** Chronic hepatitis B
- D.** NASH
- E.** Alpha-1-antitrypsin deficiency

5. Why is serum copper not used to diagnose Wilson disease?

- A.** Serum copper is always normal in Wilson disease
- B.** It may be low, normal, or elevated depending on disease stage
- C.** Copper cannot be measured in serum
- D.** Serum copper is only elevated in acute Wilson disease
- E.** Copper is only present in urine, never in serum

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — NAFLD develops in association with insulin resistance and the metabolic syndrome (type 2 DM, central obesity, dyslipidemia, hypertension).

2 • D — Hemochromatosis is confirmed by demonstrating increased tissue iron using the Prussian blue stain on liver biopsy.

3 • C — Wilson disease is caused by loss-of-function mutation of ATP7B, impairing copper excretion into bile and incorporation into ceruloplasmin.

4 • E — PAS-positive, diastase-resistant globules in hepatocytes are the classic liver biopsy finding in alpha-1-antitrypsin deficiency, which also causes emphysema via unopposed protease activity in the lung.

5 • B — Serum copper levels are of no diagnostic value in Wilson disease because they may be low, normal, or elevated depending on the stage of liver disease; low ceruloplasmin and elevated hepatic/urinary copper are used instead.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL4 (titled "Liver, pancreas and biliary system" — slide content covers liver only), Dr Savan Saeed Azeez, Hawler Medical University
Study alongside the lecture, not instead of it.